Introduction

In the earlier chapters we have learnt to construct probability distributions of certain Random variables. The probability of the various values of the random variables are obtained in accordance with the events and the nature of the experiment

In this chapter we are going to see some distributions called theoretical distributions. In these distributions, probabilities of the events are to be obtained using (formula) derived under certain conditions or assumptions. Of the many distributions available, the more common are Bernoulli, Binomial, Poisson, Normal, and Uniform distributions.

In practical situations one has to thoroughly understand the random environment and to describe it. It is followed by suggesting one of the above probability functions suitable to the situation and to obtain the requirement. The characteristics of the probability distributions such as Central Tendency, Dispersions, and Skewness are also to be studied.

10.1 Discrete distributions

10.1.1 Bernoulli's Distribution

It is discovered by a Swiss Mathematician James Bernoulli (1654-1705) for a trial which has only two outcomes viz. a success with probability p and a failure with probability $q = 1 - p$.

Definition

A random variable X is said to follow a Bernoulli distribution if its probability mass function is given by

$$P(X = x) = \begin{cases} p^x q^{1-x} & x = 0, 1 \\ 0 & otherwise \end{cases}$$

Characteristics of Bernoulli Distribution

(i) Number of trials is one

(ii) $q = 1 - p$

(iii) Constants of the distributions

(iv) (i) mean = p (ii) variance = pq (iii) standard deviation = $\sqrt{pq}$